

Learning inter-family correlations with a shared hidden layer: a two-family Ising model as a controlled test case

Abstract

We consider a toy statistical-mechanics system made of two weakly coupled Ising subsystems (“families”), each with its own sparse random intra-family couplings and a sparse set of inter-family couplings. We ask whether a Restricted Boltzmann Machine (RBM), trained only on family-resolved data, can be extended with a small number of additional hidden units to recover the inter-family coupling structure without disturbing what has already been learned about each family in isolation. The extension is built by freezing the parameters of two independently trained RBMs inside a larger, paired RBM and leaving only a block of newly added hidden units, connected to the full joint visible layer, free to adapt. These added units recover genuine cross-family structure: the correlation matrix sampled from the trained paired model at equilibrium correlates with the true coupling matrix at $r \approx 0.84$, close to the finite-sample ceiling set by the data themselves ($r \approx 0.85$), and far above the $r \approx 0$ obtained from two independent, uncoupled RBMs. We then scan the number of added units, trained long enough at a low enough learning rate to reach full convergence at every size, and find a less comfortable picture than a shorter, less careful sweep first suggested: cross-family recovery saturates once roughly 20–30 units are present, but each family’s own single-site statistics take a corresponding, persistent hit once the added units start doing genuine cross-family work – one that does *not* shrink again as more units are added, unlike what under-trained large-capacity runs at a higher learning rate had appeared to show.

1 Motivation

Many complex systems are not single homogeneous systems but collections of weakly interacting subsystems: coupled spin lattices, interacting neural populations, or correlated biological sequence families. A recurring question in this setting is whether a genuine cross-subsystem coupling can be identified from data, as opposed to a spurious correlation induced by imperfectly estimated single-subsystem statistics. We study this on a toy model where the answer is known by construction: two Ising subsystems joined by a sparse matrix of inter-family couplings. Rather than training a single RBM on the joint data, we train two independent RBMs on each family alone, then add a small block of *additional* hidden units on top of the two frozen models to account for whatever they fail to explain – the inter-family correlations. This isolates, quite literally, the channel through which the model can represent cross-family information, so that channel’s output can be checked directly against the known ground truth.

2 Model and data

Each family $X \in \{A, B\}$ consists of $N = 30$ Ising spins $s_i^X = \pm 1$. The joint system of 60 spins is governed by

$$\mathcal{H}(\mathbf{s}_A, \mathbf{s}_B) = - \sum_{i < j} J_{ij}^A s_i^A s_j^A - \sum_{i < j} J_{ij}^B s_i^B s_j^B - \sum_{i,j} K_{ij} s_i^A s_j^B - \sum_i h_i^A s_i^A - \sum_i h_i^B s_i^B, \quad (1)$$

sampled at inverse temperature $\beta = 0.5$ by single-spin-flip Metropolis–Hastings, with 5000 sweeps of burn-in and 100 sweeps of thinning between the 20 000 recorded configurations (70/15/15 train/validation/test split). The intra-family couplings J^A, J^B sit on sparse random graphs (edge density 0.12, $J \in [-0.8, 0.8]$); the inter-family coupling K is sparser still, 100 non-zero entries out of 900 possible pairs (11% density), $K \in [-0.9, 0.9]$. The local fields are drawn from $h \in [-0.3, 0.3]$ – wide enough that each site’s marginal $\langle s_i \rangle$ carries a signal well above the sampling noise of the diagnostics used below (Section 6 discusses why this matters). No information about J^A, J^B or K is passed to the learning stage.

3 RBM architecture and the paired construction

We use a Potts RBM: visible units are one-hot, $q = 2$ -state variables (the natural generalisation of an Ising spin), paired with hidden units of the *xReLU* family (a continuous, dReLU-type response with unbounded asymmetry). Training is by Persistent Contrastive Divergence – persistent Monte Carlo chains advanced 50 sweeps between successive Adam updates ($\beta_1 = 0$, $\beta_2 = 0.999$, learning rate 10^{-4}), batch size 512 – for 20 000 steps per private RBM (30 hidden units each).

The *paired* model (Fig. 1) concatenates the two families’ visible layers (60 spins) and the two private hidden layers, plus H_{add} newly added hidden units. The private A – A and B – B weight blocks are initialised from the independently trained RBMs and then *frozen*: after every gradient step, they are reset to their reference values, discarding whatever update the optimiser proposed. The cross blocks (A ’s hidden units to B ’s visible units, and vice versa) are pinned at exactly zero throughout. The only free parameters are the added units’ weights to the full 60-spin visible layer and their own biases – trained for a further 10 000 steps, with the raw gradient’s global norm clipped to 1 before every Adam update (Section 5 discusses why this is necessary at this learning rate and step count). By construction, any cross-family dependence the paired model can represent must flow through these added units.

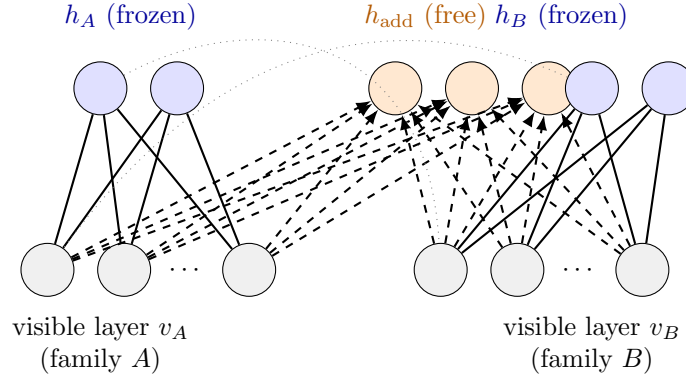


Figure 1: Architecture of the paired RBM. Solid: private A – A / B – B blocks, frozen at the independently trained values. Dotted: cross blocks h_A – v_B , h_B – v_A , pinned at zero. Dashed: the only free parameters, connecting the added units to the full visible layer.

4 Diagnostics

Training-time diagnostics compare data- and model-averaged moments along the same short persistent chains used for the gradient – cheap, but not a substitute for equilibrium, since those chains are not guaranteed to be mixed. The decisive test is performed after training: 5000 independent Gibbs chains, each run for 5000 sweeps from the visible layer’s own marginal,

used to estimate the model’s cross-family correlation $\langle s_i^A s_j^B \rangle$ and compare it against the true K , both by the fraction of the 100 strongest true couplings recovered among the model’s own 100 strongest (“top- K overlap”) and by the whole-matrix Pearson correlation with K . Both are also computed for the training data themselves (a finite-sample ceiling, since K is sparse enough that even a perfect model cannot reach $r = 1$ against it) and for a null model of two independent, uncoupled RBMs (zero learned coupling).

5 Results: $H_{\text{add}} = 30$

We first look in detail at one representative configuration, $H_{\text{add}} = 30$ with training seed 5 (chosen, among the 10 seeds run at this H_{add} , for a particularly clean and legible pseudo-likelihood trace), before scanning H_{add} itself in Section 6.

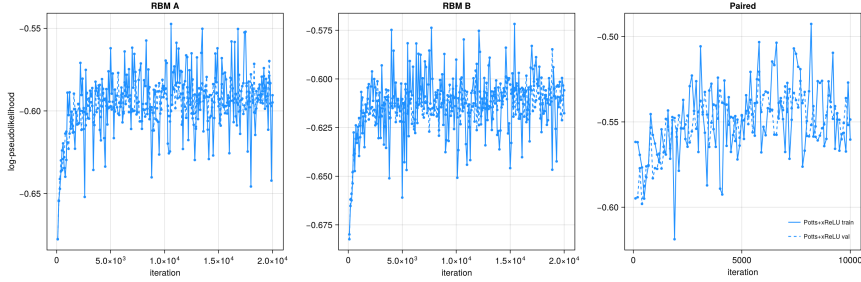


Figure 2: Training/validation pseudo-likelihood of the private RBMs and the paired RBM, vs. gradient step.

Pseudo-likelihood rises steadily and tracks its validation value closely for both families and for the paired model (no overfitting; Fig. 2). Fig. 3 shows the corresponding training-time hidden-unit activation alignment: both private RBMs stay close to $r \approx 0.95$ throughout, while the added units’ alignment with their own (short, unequilibrated) chains *improves* over the course of paired training, from $r \approx 0.90$ to $r \approx 0.995$. The complementary check on single-site visible marginals (Fig. 4) tells the opposite story for the paired model: its $\langle v \rangle$ alignment *degrades* as training proceeds, from $r \approx 0.85$ to $r \approx 0.3$ – 0.5 , even as the private RBMs’ own $\langle v \rangle$ alignment stays flat around $r \approx 0.75$ – 0.85 throughout. This is the training-time signature of the same collateral effect quantified at equilibrium below and studied systematically in Section 6.

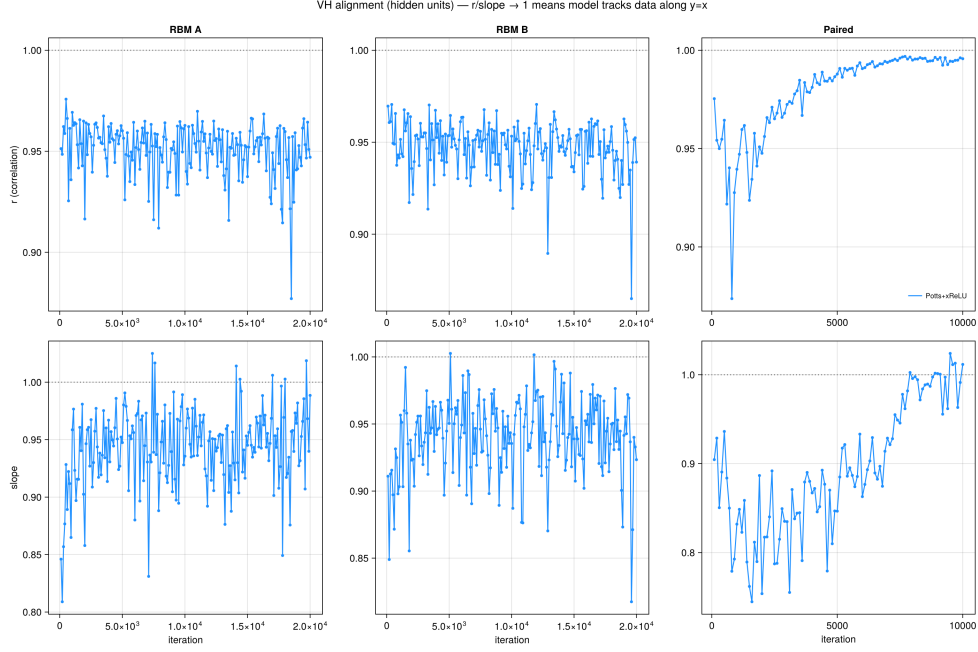


Figure 3: Training-time alignment (correlation r and regression slope, data vs. model) of hidden-unit activations $\langle hv \rangle$, vs. paired-training step for the added units (right column) and gradient step for the private RBMs (left, middle columns).

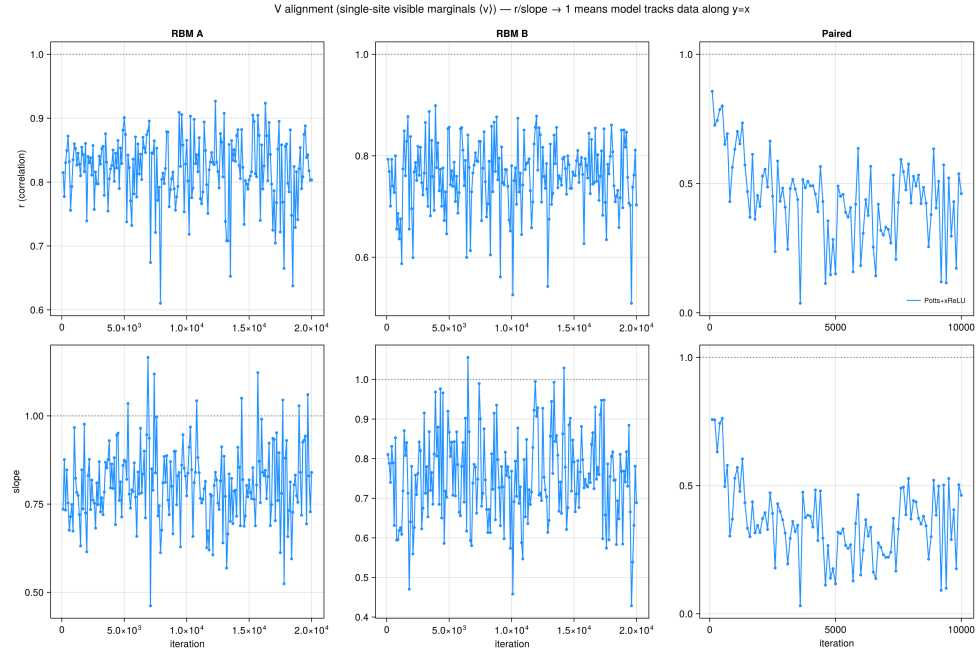


Figure 4: Training-time alignment of single-site visible marginals $\langle v \rangle$, same layout as Fig. 3. The paired model's own alignment (right column) degrades as training proceeds, opposite to Fig. 3.

The freeze check confirms the private blocks are untouched at every logged step of paired training. The training-time interfamily correlation check ($\langle v_i^A v_j^B \rangle$, data vs. model, Fig. 5 left) climbs from $r \approx 0.25$ to a stable $r \approx 0.80$ over the 10000 paired steps – encouraging, but

measured on short, unequilibrated chains and not yet the decisive test.

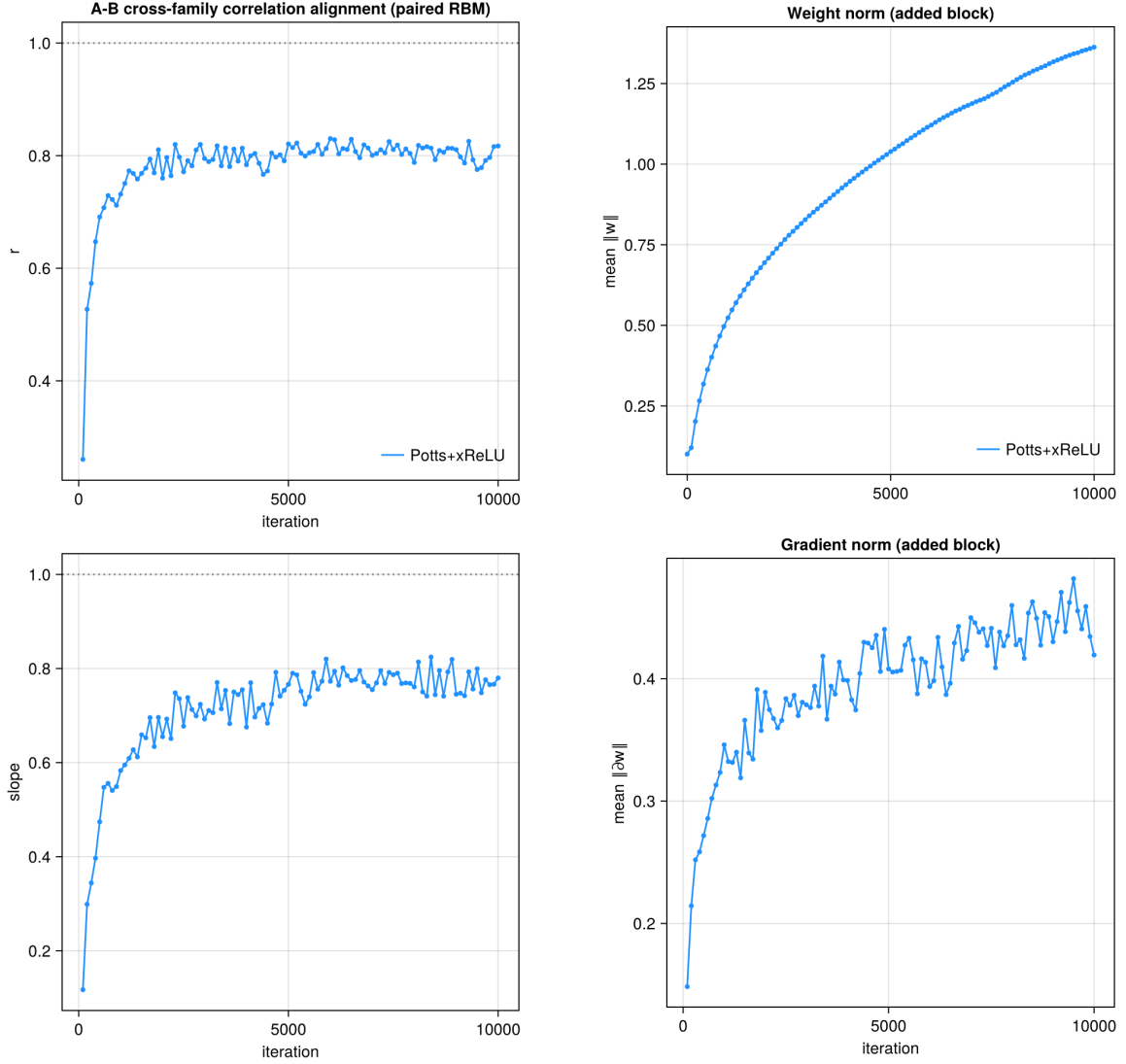


Figure 5: Left: training-time A – B cross-family correlation alignment of the paired model. Right: mean weight norm and gradient norm of the added-unit block over paired training. With gradient-norm clipping (Section 3), both grow smoothly and saturate; without it, the same 10 000-step, learning-rate- 10^{-4} combination produced a late-training runaway in 54 of the 80 swept configurations (weight norm mean $2.3 \rightarrow 14.4$, max $4.3 \rightarrow 95.9$, over the last ~ 900 steps, with pseudo-likelihood collapsing in step) – a single large gradient spike late in training, which the slow-moving ($\beta_2 = 0.999$) second-moment estimate cannot absorb quickly enough. The affected runs were archived and all were successfully retrained with clipping; every one of the 80 configurations reported here is clipping-stabilised and collapse-free.

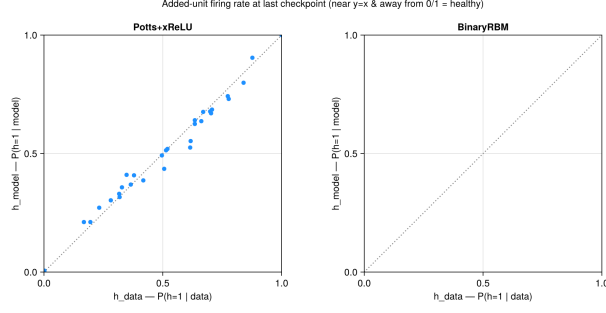


Figure 6: Added units’ firing rate $P(h = 1)$ at the final training checkpoint, data vs. model – close to $y = x$ and away from the 0/1 extremes, indicating healthy, non-degenerate activations.

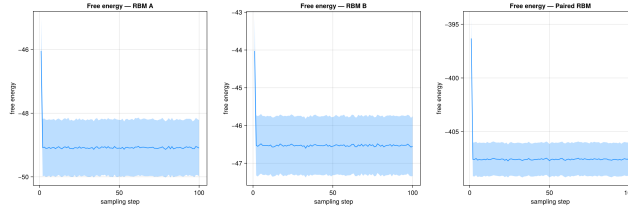


Figure 7: Free energy of the Gibbs chains vs. sweep count, averaged over 5000 chains. A plateau is reached well inside the 5000-sweep burn-in used for every equilibrium result below.

The decisive test draws long, properly burnt-in Gibbs chains (Fig. 7 confirms equilibration well within budget). Fig. 8 decomposes the resulting equilibrium samples by hidden block: the frozen blocks reproduce the data well ($\rho = 0.918$ for both $\langle h_A v_A \rangle$ and $\langle h_B v_B \rangle$), while the added units’ correlation with the *full* visible layer tracks the data almost exactly ($\rho = 0.996$ for $\langle h_{\text{add}} v_{\text{full}} \rangle$).

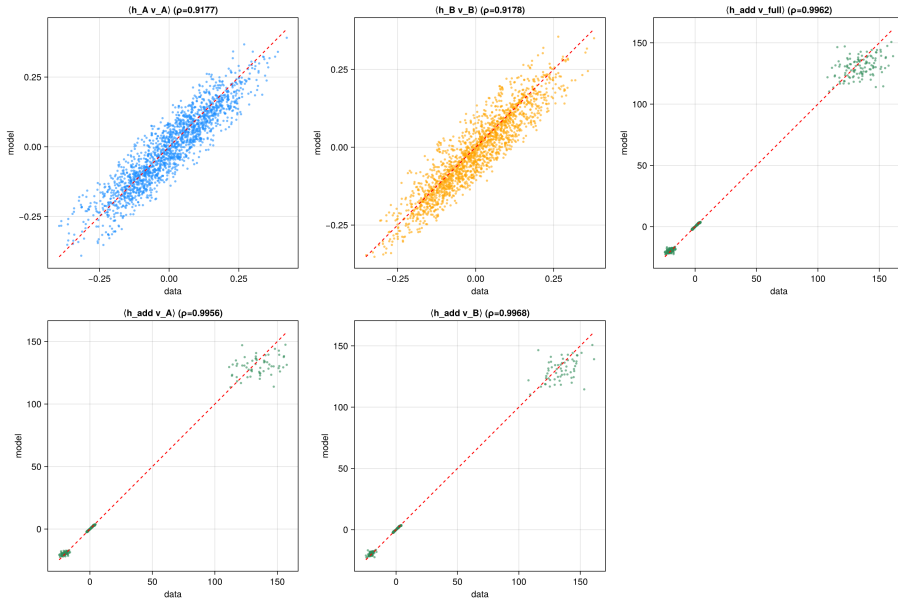


Figure 8: Data- vs. model-side $\langle hv \rangle$ from equilibrium Gibbs samples, split by hidden block: frozen A – A / B – B blocks, and the added units against the full visible layer.

The central result is the direct comparison, at equilibrium, between the recovered and true

cross-family correlation matrices (Fig. 9, Table 1): the paired model recovers 67% of the 100 strongest true couplings among its own 100 strongest, against a 72% ceiling from the finite-sample data and 10% for the independent-RBM null; the whole-matrix correlation with K reaches $r = 0.837$ against a data ceiling of $r = 0.845$ and an independent baseline consistent with zero. Family-level intra-family correlation is essentially preserved under pairing (off-diagonal $|\text{corr}|$: family A $0.0565 \rightarrow 0.0606$, family B $0.0455 \rightarrow 0.0503$).

Table 1: Equilibrium validation against the true K ($H_{\text{add}} = 30$, seed 5; 5000 Gibbs chains, 5000-sweep burn-in).

	paired model	data (ceiling)	independent (null)
top-100 overlap	0.67	0.72	0.10
corr. with K	0.837	0.845	-0.027

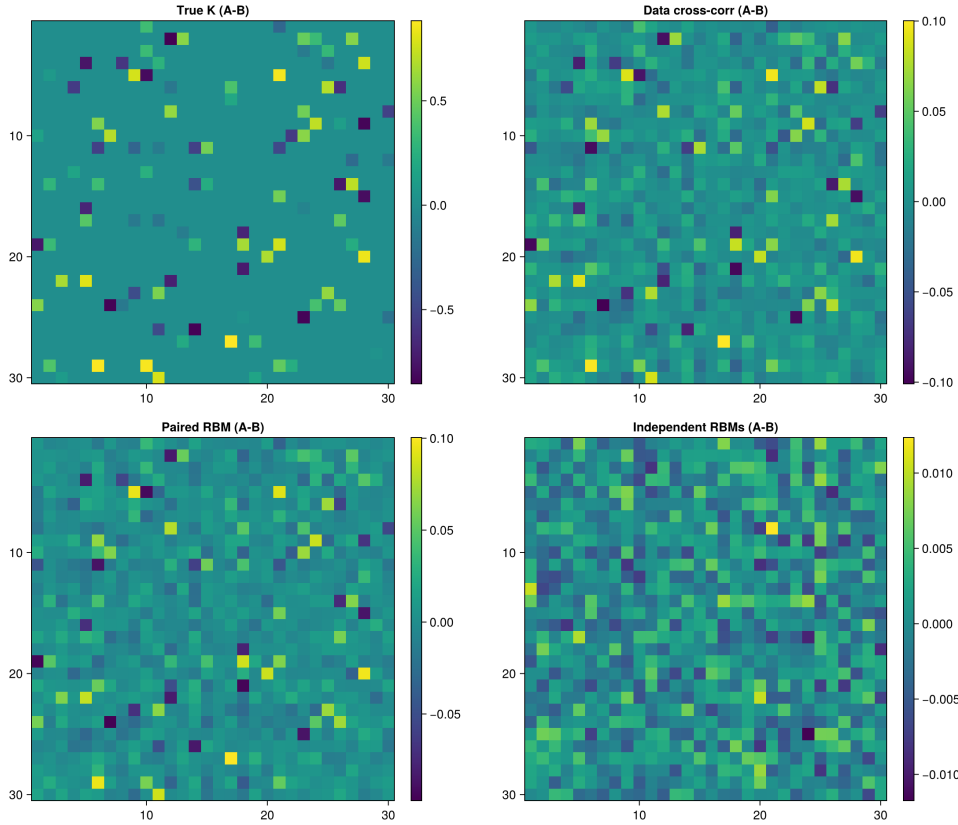


Figure 9: Cross-family correlation matrices at equilibrium: true K , empirical data correlation, correlation recovered by the paired model, and correlation from two independent, uncoupled RBMs. The paired model reproduces the sparse support of K that is absent from the independent baseline.

6 How many added units are needed? A capacity sweep

$H_{\text{add}} = 30$ above was a fixed, unjustified choice. Since the added units are the sole channel for cross-family information, how many are actually needed, and is there a cost – not to recovery, but elsewhere – to using more or fewer?

Why the wide field range. Answering this needs a diagnostic finer than pairwise correlation: the single-site marginal $\langle v_i \rangle$. Under a narrow field range ($h \in [-0.05, 0.05]$, tried first), the genuine site-to-site variation in $\langle v_i \rangle$ has standard deviation ≈ 0.008 , smaller than the sampling error of a mean over even 1000 equilibrium samples (≈ 0.016) – an observable that is structurally unusable regardless of model quality. Widening the field to $h \in [-0.3, 0.3]$ (Section 2) raises the genuine signal to ≈ 0.045 , comfortably above this noise floor, without touching the coupling structure under study.

Two distinct failure modes. With the wider field, the private RBMs’ $\langle v_i \rangle$ agreement is excellent ($\rho \approx 0.99$) and behaves like a noise-limited quantity. The *paired* model’s own $\langle v_i \rangle$, however, does not: at $H_{\text{add}} = 30$ it averages $\rho = 0.263$ with a standard deviation of only 0.044 across 10 independently trained seeds – each seed drawing its own weight initialisation, mini-batch order, and 5000-sweep equilibrium chains from scratch. A quantity this reproducible across fully independent runs cannot be sampling noise; it is a real, stable mismatch. The reason is structural: only the added units can move a site’s marginal away from what its frozen private block predicts, but a hidden unit cannot encode a correlation between two sites without also nudging each site’s own marginal – the two effects share the same weight and are not separable.

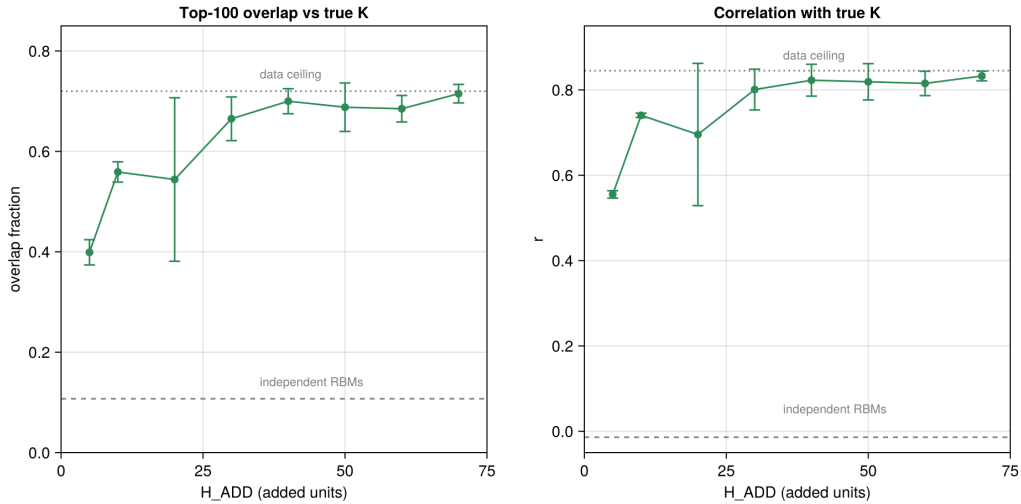


Figure 10: Cross-family recovery vs. H_{add} (equilibrium Gibbs samples, $N = 5000$; mean $\pm$ standard deviation over 10 independent training seeds per point). Both metrics rise steeply up to $H_{\text{add}} \approx 30$ –40 and then plateau, within the run-to-run spread, to 70.

The sweep. Repeating the full pipeline at $H_{\text{add}} \in \{5, 10, 20, 30, 40, 50, 60, 70\}$, each with 10 independent random seeds (governing weight initialisation, minibatch order and the persistent/equilibrium chains), gives 80 independent runs (Fig. 10). Recovery rises steeply from $H_{\text{add}} = 5$ (top-100 overlap 0.40 ± 0.03) through $H_{\text{add}} = 20$ (overlap 0.54 ± 0.16 – the one point with markedly higher seed-to-seed spread, discussed below) to $H_{\text{add}} \approx 40$ –70 (overlap 0.68 – 0.72 , standard deviation ≤ 0.05 , matching the 0.72 data ceiling within noise): thirty-to-forty added units already recover essentially everything extractable from this coupling matrix, and the small seed-to-seed spread from $H_{\text{add}} = 30$ on confirms the plateau is real, not a single lucky run. At $H_{\text{add}} = 20$ specifically, 4 of the 10 seeds land on a markedly worse optimum (overlap 0.26 – 0.49) while the other 6 already reach 0.63 – 0.69 – consistent with this being the narrowest part of the capacity transition, where whether a given run finds the good solution is more a matter of chance than at either lower or higher H_{add} .

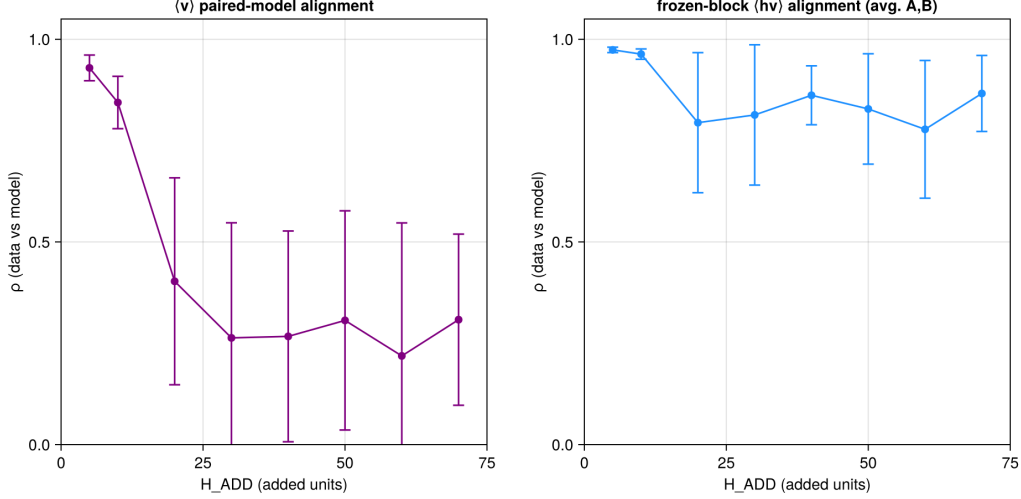


Figure 11: Collateral effect on single-site statistics, same 80 runs: paired model’s $\langle v \rangle$ agreement (left) and frozen-block $\langle hv \rangle$ agreement, averaged over A, B (right). Points are the mean over 10 seeds per H_{add} ; error bars are the standard deviation.

The collateral cost on single-site statistics (Fig. 11) tells a monotone, not U-shaped, story here: agreement is high while H_{add} is still too small to do much cross-family work ($H_{\text{add}} = 5$, mean $\langle v \rangle$ agreement 0.93; $H_{\text{add}} = 10$, mean 0.84), drops sharply once the added units start recovering real structure ($H_{\text{add}} = 20$, mean 0.40), and then sits on a noisy plateau around 0.22–0.31 all the way out to $H_{\text{add}} = 70$, with no sign of recovery at the largest sizes tested. The frozen-block $\langle hv \rangle$ agreement (right panel) shows the same shape with a shallower dip (from ≈ 0.97 at $H_{\text{add}} \leq 10$ to a 0.78–0.87 plateau beyond it). This contradicts what an earlier, shorter sweep at a ten-fold higher learning rate and a third of the paired-training steps had suggested: there, the same metric appeared to recover part-way back once H_{add} was pushed well past the point needed for recovery. Training every H_{add} here for the same 10 000 steps at $\text{lr} = 10^{-4}$ (long enough, with gradient clipping, to reach a stable plateau at every size – Fig. 5) removes that apparent recovery: it was most likely an artefact of the largest- H_{add} paired models being relatively under-trained within a fixed, short step budget, not a genuine benefit of spreading the same workload across more units.

Interpretation. The collateral shift is not a transient, undertraining artefact but a persistent cost of enabling any real cross-family learning: below the capacity needed for recovery, the added units simply are not doing much, so they cannot distort much either; once they start doing genuine work, each family’s own single-site statistics take a corresponding, lasting hit that further over-provisioning does not undo once training is run to full convergence. The practical lesson is more sobering than a simple “add more units” fix: recovering cross-family structure and keeping each subsystem’s own statistics undistorted are in real, structural tension in this construction, and no amount of added-unit capacity resolves it on its own.

7 Discussion

A small, dedicated block of hidden units, added on top of two frozen, independently trained models, recovers most of an otherwise hidden inter-family coupling, confirmed directly against ground truth through equilibrium sampling rather than training curves alone. The natural whole-matrix correlation against a sparse K is structurally capped well below 1 even for a perfect model (Table 1’s 0.845 ceiling); reporting against that ceiling, not against $r = 1$, is

essential. The capacity sweep (Section 6), run long enough and with gradient clipping to reach full, stable convergence at every size, revises the earlier picture from a shorter, higher-learning-rate sweep: the tension between cross-family recovery and undistorted single-site statistics does not resolve itself once enough capacity is added – it is structural, tracking how much genuine cross-family work the added units are doing rather than how many of them there are. A further methodological note follows from the same careful rerun: at this learning rate and step count, unclipped gradients produced a late-training instability in 54 of the 80 swept configurations, visible only as a sharp, late collapse in pseudo-likelihood and a runaway added-unit weight norm – a reminder that training-time diagnostics need to be checked for every configuration in a sweep, not spot-checked on a few, before their downstream conclusions are trusted.